

Bridging Embodied Artificial Intelligence and Accessibility

Proposed Full-Day Workshop at HRI 2027

Hamza El Alaoui[✉]
Carnegie Mellon University
Pittsburgh, PA, USA

Harish Ram Nambiappan
Texas A&M University
Corpus Christi, TX, USA

Atieh Taheri
Carnegie Mellon University
Pittsburgh, PA, USA

Chen Liang
University of Michigan
Ann Arbor, MI, USA

Max Pascher
TU Dortmund University
Dortmund, Germany

Vinitha Ranganeni
Stanford University
Stanford, CA, USA

Stephanie Valencia
University of Maryland
College Park, MD, USA

Zackory Erickson
Carnegie Mellon University
Pittsburgh, PA, USA

Tapomayukh Bhattacharjee
Cornell University
Ithaca, NY, USA

Anhong Guo
University of Michigan
Ann Arbor, MI, USA

Eshed Ohn-Bar
Boston University
Boston, MA, USA

Jeffrey P. Bigham
Apple Inc. & Carnegie Mellon
Pittsburgh, PA, USA

Abstract

Assistive robots and embodied AI create new opportunities to support people with disabilities in physical assistance, communication, navigation, rehabilitation, and everyday life. Their usefulness depends on how people express goals, share control, understand robot behavior, and recover from failures across diverse abilities and contexts. This workshop brings together researchers and practitioners in human–robot interaction, accessibility, robotics, AI, rehabilitation, wearables, and extended reality, alongside people with lived experience of disability. We will examine how to design assistive human–robot interaction that is usable, trustworthy, adaptable, and centered on user agency. Topics include physical assistance and caregiving; accessible teleoperation and shared autonomy; speech and multimodal control; visual access and navigation; XR interfaces for robot interaction; and disability-led design and evaluation. Through paper presentations, demonstrations, a panel, small-group discussions, and collaborative synthesis, participants will compare methods and identify research questions that connect technical capabilities to people’s everyday priorities. The proposed full-day workshop will produce a public discussion summary, use-case scenarios, and a shared research agenda for accessible HRI, with opportunities for continued collaboration.

 [EmbodiedAccess.github.io](https://github.com/EmbodiedAccess)  helalaou@cs.cmu.edu

CCS Concepts

• **Human-centered computing** → **Accessibility**; *Interaction techniques*; • **Computer systems organization** → **Robotics**.

Keywords

Assistive robotics, accessibility, human–robot interaction, embodied AI, shared autonomy, multimodal interaction

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1 Motivation and Scope

Assistive robots can support everyday activities, yet useful assistance requires more than capable autonomy: people must be able to direct, understand, and reshape that assistance [32]. Accessible teleoperation and in-home use illustrate the importance of adapting control to individual abilities [34, 43]. Our central question is: *How can assistive robots expand people’s capabilities while preserving their agency, preferred forms of assistance, and meaningful control?*

The workshop connects technical HRI research with accessibility and disability-led design. In keeping with HRI 2027’s “Innovative HRI” theme, we treat innovation as advances in both robot interaction and the methods used to define and evaluate assistance. We address six complementary themes:

- **Physical assistance and caregiving:** manipulation and activities of daily living [6, 32], including robot-assisted feeding and inside-mouth bite transfer [13, 21]. Wearable grasp assistance [60] and simulation of physical assistance [11] broaden the design space. We will examine comfort, consent, safe recovery, and the structural and social requirements of everyday assistance [36].
- **Accessible control and shared autonomy:** body-machine interfaces and customizable controls [20, 43, 44]; shared-control frameworks [1, 12]; and human-in-the-loop adaptation and low-dimensional action spaces [15, 25]. Adaptive input mappings [37, 41], intent communication [38], and evaluations of alternative inputs and in-home control [34, 42] motivate discussion of intervention, handoff, and personalization that preserve meaningful user control.
- **Speech and multimodal interaction:** interaction proxies [23, 24] and language-supported interfaces [8, 9, 35, 59]; accessible speech and multimodal systems [28, 29, 31]; and combined gaze and speech [22]. Work on senior care [45], wheelchair interfaces [47], affective touch [33], and mouth gestures for feeding [7] broadens the modalities considered.

We will also discuss embodied communication and shared object awareness [17, 46, 56].

- **Visual access and navigation:** collision-avoidance interfaces [30], access to physical interfaces [16], and context-aware descriptions [3–5] for blind and low-vision (BLV) people. Studies of guide robots [48] and language-guided navigation [19] inform accessible navigation. We will consider how accessible feedback and spatial information support action in changing environments, alongside advances in autonomous mobility [10].
- **XR for robot interaction:** virtual design elements [58], adaptive virtual bodies and overlays [39, 40], and augmented-reality support for manipulation and motion-intent communication [2, 38, 57]. Transitional XR [37], hands-free controllers [53, 54], and studies of virtual embodiment [49] inform accessible prototyping and evaluation of shared control.
- **Disability-led design and evaluation:** co-design and inquiry grounded in disability studies [18, 26, 27, 56], including critical reflection on paternalism and autonomy [14]. Lived-experience research [52], physical agency and felt experience [51], authorship in AI-assisted communication [55], and agency in AI-mediated guidance [50] motivate success measures beyond task efficiency for deployment in homes and communities.

2 Format and Program

We propose an in-person, full-day workshop on March 8, 2027, for approximately 25–50 participants, subject to conference acceptance and scheduling. At least one organizer will attend in person. A full day allows participants to encounter systems and perspectives through talks and demos, examine cross-cutting tensions in a panel, and then develop shared questions through facilitated breakouts and synthesis. These connected activities require time for accessible participation and exchange across disciplines.

09:00–09:20 Welcome, access notes, and introductions.

09:20–09:50 Keynote framing talk and discussion.

09:50–11:00 Paper session I: up to seven contributions.

11:00–12:00 Demonstrations and posters.

12:00–13:15 Lunch break.

13:15–14:15 Paper session II: up to six contributions.

14:15–15:00 Panel on agency, safety, and shared control.

15:00–16:00 Facilitated thematic small-group discussions.

16:00–16:40 Group reports and collaboration pitches.

16:40–17:00 Synthesis, commitments, and next steps.

Short breaks will be accommodated within session blocks. Each breakout will use a shared prompt: identify a use case, a control or access barrier, an interaction approach, and a way to evaluate it with intended users. An organizer will facilitate each group and invite written as well as spoken contributions.

3 Audience, Recruitment, and Participation

We welcome researchers, practitioners, designers, students, clinicians, and people with lived experience of disability across HRI, accessibility, robotics, AI, rehabilitation, wearables, and XR. No particular robot platform or technical prerequisite is required. We will circulate a call through the organizers’ research groups, HRI

and accessibility mailing lists, professional networks, and relevant community contacts, inviting both established and emerging perspectives.

Participants may propose short or long papers, position statements, or system/demo descriptions. The call will specify submission lengths, dates, and the submission route after workshop acceptance. Organizers will review contributions for relevance, clarity, and potential to stimulate discussion, seeking complementary methods and perspectives. An interest form will collect topics, demo plans, and optional access requests; participants may also contact the organizers directly. We will share accepted materials in advance with authors’ permission.

We will provide readable digital materials and written activity prompts, invite access requests privately, and coordinate room and access arrangements with the conference. Demonstrations will include verbal descriptions and seated or non-hands-on participation options. Participation in photographs or recordings will be optional. No conference-supported hybrid provision is assumed.

4 Documentation and Outcomes

Each group will record its use case, open questions, evaluation priorities, and possible collaborations in a shared template. Organizers will synthesize these contributions into a discussion summary and research agenda for accessible HRI, and invite participant review before public release. The workshop website, <https://embodiedaccess.github.io>, will host consented papers, demo links, resources, and outcomes. Contributor approval will govern attribution and sharing; private access requests will not be published. We will prepare HRI’s one-page archival workshop summary if accepted; participant papers will not automatically enter ACM proceedings.

5 Organizers

Hamza El Alaoui is a Ph.D. student at CMU's School of Computer Science. His research focuses on cyber-physical human augmentation, combining machine learning, embedded systems, robotics, and HCI to create systems that extend human communication, perception, and physical capabilities. His work spans language technologies, multimodal sensing, human-AI integration, and accessibility systems. He previously worked on machine learning at Oracle and was a visiting scientist at Sony Computer Science Laboratories.

Harish Ram Nambiappan is an Assistant Professor at Texas A&M University. He received his Ph.D. in Computer Science from The University of Texas at Arlington. His research focuses on developing assistive navigation and robotic systems that enable BLV people to navigate dynamic indoor environments and interact directly with robots. He has built cross-modal interfaces that allow users to locate objects, direct robotic manipulation, and remain safe around moving robots and people. His broader work also includes autonomous service robots for healthcare, placing his research at the intersection of robotics, accessibility, autonomy, and HRI.

Atieh Taheri is a Presidential Postdoctoral Fellow at CMU. Her research combines accessibility, human-centered AI, mixed reality, and participatory design to build interactive systems that expand agency and creative expression for people with disabilities. Her work includes hands-free controllers, tactile interfaces, accessible conversational systems, VR experiences, and accessible generative-AI tools. She previously conducted research at Google and Apple, and earned her Ph.D. in Electrical & Computer Engineering from the University of California, Santa Barbara.

Chen Liang is a Ph.D. candidate in Computer Science & Engineering at the University of Michigan. His research develops human-AI systems that make UIs more intelligent, adaptable, and accessible. He creates AI-powered interaction proxies that translate or automate actions on users' behalf without requiring modifications to the underlying interface. He previously conducted research at Adobe.

Max Pascher is a Postdoctoral Researcher at TU Dortmund University. His research focuses on AI-enhanced human-robot collaboration, including shared-control techniques for assistive robotic arms, intervention strategies for autonomous robotic tasks, and XR-based tools for designing and evaluating human-robot interfaces. He also works on multimodal interaction and methods that keep users in control of autonomous robotic systems. He received his Ph.D. in Human-Robot Interaction from the University of Duisburg-Essen.

Vinitha Ranganeni is a Postdoctoral Scholar at Stanford University. Previously, she was the Human-Robot Interaction Lead at Hello Robot, where she led the development and real-world evaluation of human-centered software for the Stretch mobile manipulator. Her work focuses on accessible assistive robotics for people with disabilities, including customizable interfaces and in-home deployments that support greater independence. She received her Ph.D. in Computer Science from the University of Washington.

Stephanie Valencia is an Assistant Professor of Information Science at the University of Maryland. Her research focuses on equitable access to assistive technology, open-source hardware, and the design of communication systems that increase user agency and accessibility. Using participatory design, she develops AI-enabled and embodied expressive technologies for augmentative and alternative communication (AAC) users, with an emphasis on building, deploying, and rigorously evaluating systems in real-world settings. She received her Ph.D. in HCI from CMU.

Zackory Erickson is an Assistant Professor at CMU's Robotics Institute and director of the RCHI Lab. His research develops robot-learning, mobile-manipulation, sensing, and control methods for physical HRI to help older adults and people with disabilities live independently. His work spans assistance with feeding, dressing, bathing, and personal care; wearable interfaces for limited hand function; simulation; haptic perception; and safe, personalized robot autonomy. He received his Ph.D. in Robotics from Georgia Tech.

Tapomayukh Bhattacharjee is an Assistant Professor of Computer Science at Cornell University, where he directs the EmPRISE Lab. His research focuses on physical robot caregiving and physical HRI, particularly assistive robots that support people with mobility limitations in activities of daily living. His lab takes a full-stack robotics approach spanning sensing, perception, planning, learning, control, manipulation, haptic perception, and tactile sensing. He received his Ph.D. in Robotics from Georgia Tech and was a Postdoctoral Researcher at the University of Washington.

Anhong Guo is an Assistant Professor of Computer Science at the University of Michigan. His research lies at the intersection of HCI, AI, and accessibility. He develops personal assistive technologies that adapt to users' individual abilities, contexts, and intentions, particularly for BLV people. His work spans human-AI systems for accessing visual information, accessible physical and touchscreen interfaces, customizable visual-assistance tools, context-aware real-world descriptions, augmented and virtual reality, and multimodal end-user programming. He received his Ph.D. in HCI from CMU.

Eshed Ohn-Bar is an Assistant Professor of Electrical & Computer Engineering at Boston University and leads the Human-to-Everything Lab. His research develops machine-intelligence systems for embodied, assistive, and autonomous applications, spanning perception, learning, planning, control, and human-machine interaction. His work includes accessible navigation and AI agents that provide real-time assistance to BLV people. He received his Ph.D. in Electrical Engineering from the University of California, San Diego, and was a Humboldt Fellow at the Max Planck Institute.

Jeffrey P. Bigham is an Associate Professor at CMU's Human-Computer Interaction Institute and Language Technologies Institute, where he directs the Beneficial Intelligence Group. He also serves as Director of Human-Centered Machine Intelligence & Responsible AI at Apple, leading research on AI/ML design, accessibility, and UI understanding. His research combines machine learning and human computation to develop reliable, human-centered AI systems for real-world use, particularly in accessibility. He received his Ph.D. in Computer Science from the University of Washington.

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